3-1-1: Networking Protocols and Services

At the end of this episode, I will be able to:

1. Identify and explain the purpose, significance and role of network protocols, services, and ports, given a scenario.

Learner Objective: *Identify and explain the purpose, significance and role of network protocols, services, and ports, given a scenario*

Description: In this episode, the learner will examine common network protocols, services and ports used in networks today. We will examine the purpose, significance and role these protocols provide.

- Trivial File Transfer Protocol (TFTP)
  - A simplified version of FTP, used for transferring small amounts of data
  - Port 69
- File Transfer Protocol (FTP)
  - Used for transferring files between a client and server on a network
  - Ports 20/21
- Secure File Transfer Protocol (SFTP)
  - An extension of SSH providing secure file transfer capabilities
  - Port 22
- Telnet
  - A protocol for bidirectional interactive text-oriented communication using a virtual terminal connection
  - Port 23
- Secure Shell (SSH)
  - Provides a secure channel over an unsecured network, often used for secure remote login
  - Port 22
- Simple Mail Transfer Protocol (SMTP)
  - The standard protocol for sending emails across the Internet
  - Port 25
- Simple Mail Transfer Protocol Secure (SMTPS)
  - A method for securing SMTP with transport layer security
  - Port 587
- Internet Message Access Protocol (IMAP)
  - Allows users to access and manage emails stored on a mail server, enabling multiple clients to access the same mailbox.
  - Port 143.
- Internet Message Access Protocol Secure (IMAPS)
  - A version of IMAP that includes encryption for secure email access and management, typically using SSL/TLS.
  - Port 993.
- Post Office Protocol 3 (POP3)
  - Designed for downloading emails from a server to a local client, with emails typically removed from the server after download.
  - Port 110.
- Post Office Protocol 3 Secure (POP3S)
  - The secure version of POP3, adding encryption (usually SSL/TLS) to enhance the security of email retrieval.
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    - Port 995.
- Domain Name System (DNS)
    - Translates domain names to IP addresses for accessing websites
    - Port 53
- Dynamic Host Configuration Protocol (DHCP)
    - Automatically assigns IP addresses and other network configurations to devices
    - Ports 67/68
- Hypertext Transfer Protocol (HTTP)
    - The foundation of data communication for the World Wide Web
    - Port 80
- Hypertext Transfer Protocol Secure (HTTPS)
    - An extension of HTTP for secure communication within a web browser
    - Port 443
- Network Time Protocol (NTP)
    - Synchronizes clocks of networked devices to Coordinated Universal Time (UTC)
    - Port 123
- Simple Network Management Protocol (SNMP)
    - Manages and monitors network devices
    - Ports 161/162
- Lightweight Directory Access Protocol (LDAP)
    - Maintains and accesses distributed directory information services
    - Port 389
- Syslog
    - A standard for message logging
    - Port 514
- Lightweight Directory Access Protocol over SSL (LDAPS)
    - LDAP over SSL/TLS for encrypted sessions
    - Port 636
- Structured Query Language (SQL) Server
    - A database server using SQL for database management
    - Port 1433
- Remote Desktop Protocol (RDP)
    - Provides a graphical interface to connect to another computer over a network
    - Port 3389
- Session Initiation Protocol (SIP)
    - Used for signaling and controlling multimedia communication sessions
    - Ports 5060/5061

- Additional Resources:
    - If applicable